

Date	20/6/2025
Team Id	LTVIP2025TMID29678
Project Name	GARAGE MANAGEMENT SYATEM
Maximum Marks	

Chapter 2: Ideation Phase

2.1 Problem Statement:

Modern vehicle service garages often struggle with manual operations, leading to inefficiencies in managing customer appointments, service records, inventory, billing, and workforce allocation. These fragmented processes result in long wait times, misplaced service histories, inventory mismatches, and poor customer satisfaction.

Additionally, garage owners lack real-time insights into business performance, technician productivity, and customer feedback, making it difficult to optimize operations or scale services. The absence of a centralized digital system also increases the risk of human error, data loss, and revenue leakage.

This project aims to develop a Garage Management System that automates and streamlines core garage functions—appointment scheduling, service tracking, inventory control, billing, and reporting—through a user-friendly digital platform. The solution will improve operational efficiency, enhance customer experience, and provide actionable analytics for business growth.

2.2 Empathy Map Canvas:



Who are we empathizing with?

- Primary User: Garage owner or manager
- Secondary Users: Technicians, customers, inventory staff



What do they say?

- "Managing appointments manually is a headache."
- "I wish I had a better way to track service history."
- "Inventory always runs out at the worst time."



What do they think?

- "Am I losing customers because of delays?"
- "I need to grow my business but don't have the tools."
- "I hope the system is easy to use and doesn't require training."



What do they feel?

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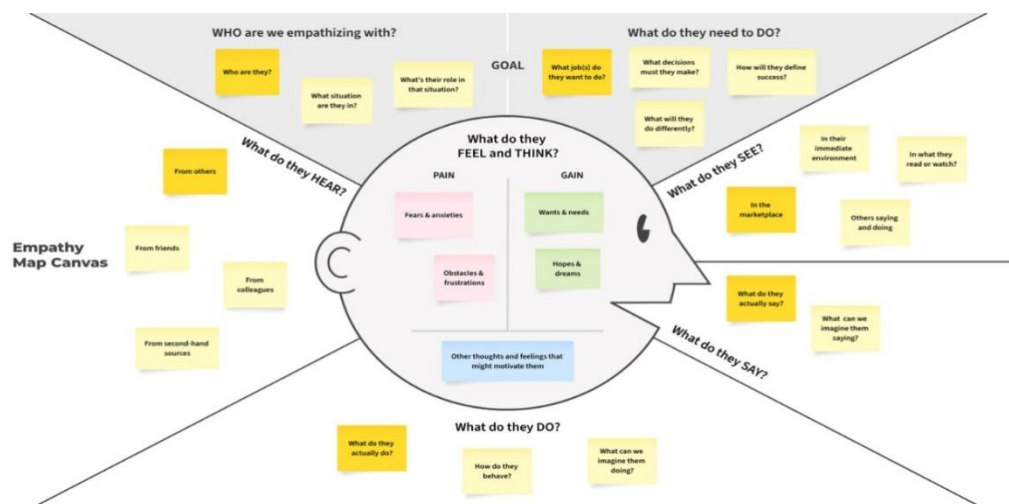
- Overwhelmed by daily operations
- Frustrated with manual paperwork
- Anxious about customer satisfaction
- Hopeful for a digital solution

What do they do?

- Schedule appointments via phone or walk-ins
- Track services using notebooks or spreadsheets
- Manually check inventory and order parts
- Handle billing and customer queries

Gains

- Streamlined operations
- Real-time inventory updates
- Improved customer satisfaction
- Data-driven decision-making



2.3 Brainstorming:

Core Functional Areas

- Appointment Scheduling: Online booking, calendar sync, SMS reminders
- Service Tracking: Vehicle history, technician notes, service status updates
- Inventory Management: Real-time stock levels, auto-reorder alerts, supplier integration
- Billing & Invoicing: Digital receipts, tax calculations, payment gateway integration

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- Customer Portal: Service history, feedback forms, loyalty rewards

📦 Innovative Add-ons

- AI Diagnostics: Predictive maintenance suggestions based on vehicle data
- QR Code Check-ins: Customers scan to log arrival and view service queue
- Live Service Feed: Customers can track progress via app or web dashboard
- Gamified Technician Dashboard: Performance badges, task timers, leaderboards
- Eco Score: Track and promote eco-friendly service practices

📋 Tech Stack Ideas

- | Layer | Tools/Tech Suggestions |
- | Frontend | React, Flutter, HTML5 |
- | Backend | Node.js, Django, Firebase |
- | Database | PostgreSQL, MongoDB, Firestore |
- | Notifications | Twilio (SMS), Firebase Cloud Messaging |
- | Hosting | AWS, Heroku, Vercel |

👤 User Personas

- Garage Owner: Wants business insights, technician performance, inventory control
- Technician: Needs task assignments, service history, time tracking
- Customer: Wants easy booking, real-time updates, transparent billing

📋 Challenges to Solve

- Manual record-keeping and lost service history
- Inventory mismatches and delays in part availability
- Long customer wait times and poor communication
- Lack of performance tracking for technicians.

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